Quiz 1

(Deadline March 19, 2026)

Department: _____

Student ID: _____

Name: _____

Problem 1

(3.5 points) Modern symmetric-key cryptography is built upon the fundamental principles of confusion and diffusion. In the Advanced Encryption Standard (AES), these principles are realized through transformations such as SubBytes (S-Box), ShiftRows, and MixColumns. Conceptually, these operations can be viewed as modern and highly structured counterparts of classical cryptographic techniques, including substitution ciphers, transposition ciphers, and matrix-based transformations such as the Hill cipher. In this quiz, we study these classical ciphers in order to develop an intuitive understanding of the core design ideas that underpin contemporary block ciphers such as AES.

Instructions: Based on the ciphertext provided below, answer the following questions. In each case, present your reasoning clearly and support your answer with an appropriate explanation.

Use the convention $A = 0, B = 1, \ldots, Z = 25$ unless otherwise specified.

a) (0.5 points) Write a program to compute the frequency distribution of the letters appearing in the given ciphertext.

b) (0.5 points) Using the standard frequency distribution of letters in English plaintext as a reference, analyze the ciphertext and attempt to break the substitution cipher.

c) (0.5 points) Let C denote the ciphertext and P denote the plaintext. Determine whether a specific mathematical relationship exists between C and P.

d) (0.5 points) Suppose the encryption function is defined as

$$f(x) = ax + b \pmod{26},$$

where x denotes a plaintext letter represented as an integer. Determine the values of a and b.

e) (0.5 points) What is the key size of a **monoalphabetic substitution cipher**? Is this key size sufficiently large to make an exhaustive-search attack impractical? Briefly explain your answer.

f) (0.5 points) For the affine substitution cipher defined by

$$f(x) = ax + b \pmod{26},$$

determine the size of its key space. Since the key space of the **affine substitution cipher** is smaller than that of the monoalphabetic substitution cipher, how might its key space be increased? **Hint:** In the class.

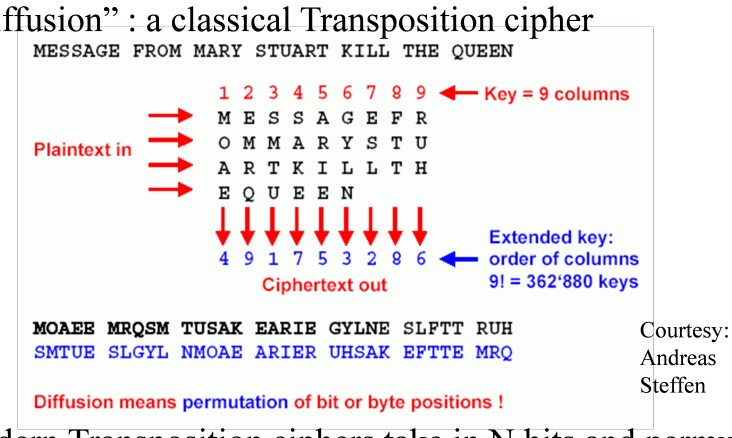
g) (0.5 points) Write a program that automatically cracks this affine cipher by trying all valid keys, scoring each decryption by English letter frequency, and printing the best candidates. Also explain each step.

Ciphertext:

```
GYOMXNOGNG QUGN ETNMX MPLMZOMXYM K TMMJOXA XEN TKZ ZMQEBMF TZEQ
KJKZQ EX YEXNMQLJKNOXA NHM TJEEF ET XMI CXEIJMFAM IHOYH
MKYH WMKZ RZOXAG IONH ON
```

Problem 2

(3 points)



Courtesy: Andreas Steffen

Try to decrypt this Columnar Transposition cipher.

Ciphertext:

`ECDTM ECAER AUOOL EDSAM MERNE NASSO DYTNR VBNLC RLTIQ LAETR IGAWE BAAEI HOR`

One possible rectangle sketch:

E	R	A
C	A	M
D	U	M
T	O	E
M	O	R
E	L	N
C	E	E
A	D	N
E	S	A

In this case we have 63 letters. Vowel frequencies can help us to determine the dimensions of the rectangle. In English approximately 40% of plaintext consists of vowels. Therefore, for the correct dimension, each row of the rectangle should be approximately 40% vowels. For example, there are 21 letters in the ciphertext. Because we know that the message completely fills the rectangle, this suggests either a 3×7 or a 7×3 array.

Statistical Analysis Using Vowel Frequency

Since English plaintext contains approximately 40% vowels, we compare the two possible rectangle dimensions.

Case 1: for 3×7 rectangle) Each row has 7 letters. Expected number of vowels per row:

$$0.4 \times 7 = 2.8$$

Row	Number of Vowels	Difference
AITMTSE	3	0.2
SRFIKOE	3	0.2
AINMLIM	3	0.2

Total difference:

$$0.2 + 0.2 + 0.2 = 0.6$$

Case 2: for 7×3 rectangle) Each row has 3 letters. Expected number of vowels per row:

$$0.4 \times 3 = 1.2$$

Row	Number of Vowels	Difference
AFL	1	0.2
SNS	0	1.2
AMO	2	0.8
III	3	1.8
RME	1	0.2
ITE	2	0.8
TKM	0	1.2

Total difference:

$$6.2$$

Since the total deviation is significantly smaller for the 3×7 rectangle, this configuration is more likely to represent the correct transposition structure.

Problem 3

(3.5 points) The plaintext "THE" is encrypted using a 3×3 Hill cipher key matrix, and the resulting ciphertext is "AJN." Please write a program to decrypt the ciphertext "AJN" back to the plaintext "THE."

In your solution, make sure to use an inverse matrix modulo 26, and explain why the inverse matrix is valid. Your program and explanation should take the following points into consideration:

1. **The determinant must be relatively prime to 26.**
   Since
   $$26 = 2 \times 13,$$
   the determinant of the key matrix must not share any common factor with 26. In other words,
   $$\gcd(\det(K), 26) = 1.$$
   Therefore, the determinant cannot be:
   - even (otherwise it shares factor 2),
   - divisible by 13.

   If this condition is not satisfied, then the matrix has no inverse modulo 26.

   Examples:
   $$\det(K) = 5 \quad \Rightarrow \quad \gcd(5, 26) = 1 \quad \text{(valid)}$$
   $$\det(K) = 12 \quad \Rightarrow \quad \gcd(12, 26) = 2 \quad \text{(not valid)}$$
   $$\det(K) = 13 \quad \Rightarrow \quad \gcd(13, 26) = 13 \quad \text{(not valid)}$$

2. **You need the determinant's modular inverse.**
   After computing $\det(K)$, you must find an integer d^{-1} such that
   $$\det(K) \cdot d^{-1} \equiv 1 \pmod{26}.$$

   Example:
   If $\det(K) = 5$, then
   $$5 \cdot 21 = 105 \equiv 1 \pmod{26},$$
   so the modular inverse of 5 modulo 26 is 21.

3. **Use the adjugate matrix, not the ordinary decimal inverse.**
   For the Hill cipher, the inverse matrix is computed as:
   $$K^{-1} \equiv (\det(K))^{-1} \cdot \operatorname{adj}(K) \pmod{26}.$$
   This is different from the ordinary real-number matrix inverse.

   Steps:
   - Compute the determinant $\det(K)$.
   - Verify that $\gcd(\det(K), 26) = 1$.
   - Compute the modular inverse $(\det(K))^{-1} \pmod{26}$.

- Compute the adjugate matrix $\operatorname{adj}(K)$.
- Multiply the adjugate matrix by the modular inverse of the determinant.
- Reduce every entry modulo 26.

4. **Reduce negative numbers modulo 26.**
   When computing cofactors or the adjugate matrix, negative numbers often appear.

   For example:
$$-1 \equiv 25 \pmod{26}$$
$$-3 \equiv 23 \pmod{26}$$

   Therefore, every entry of the matrix must be reduced into the range
$$\{0, 1, 2, \ldots, 25\}.$$

5. **The matrix must be square.**
   For the Hill cipher:

   - 2-letter blocks $\to$ use a 2×2 key matrix.
   - 3-letter blocks $\to$ use a 3×3 key matrix.

   So if ciphertext block length is 3, your key matrix must be 3×3.

In short

To find an inverse matrix in the Hill cipher, you must ensure that
$$\boxed{\gcd(\det(K), 26) = 1}$$
If this condition fails, the inverse matrix does **not** exist.

And then use:
$$\boxed{K^{-1} \equiv (\det(K))^{-1} \operatorname{adj}(K) \pmod{26}}$$

Encrypting "THE" using the 3×3 Hill Cipher

Using the convention $A = 0, \ldots, Z = 25$:
$$T = 19, \quad H = 7, \quad E = 4.$$
The Matrix Multiplication (using our key matrix K):
$$K = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix} \begin{pmatrix} 19 \\ 7 \\ 4 \end{pmatrix}$$

Row 1: $(6 \cdot 19) + (24 \cdot 7) + (1 \cdot 4) = 114 + 168 + 4 = 286$
Row 2: $(13 \cdot 19) + (16 \cdot 7) + (10 \cdot 4) = 247 + 112 + 40 = 399$
Row 3: $(20 \cdot 19) + (17 \cdot 7) + (15 \cdot 4) = 380 + 119 + 60 = 559$

Apply Modulo 26:
$$286 \equiv 0\ (A), \quad 399 \equiv 9\ (J), \quad 559 \equiv 13\ (N) \pmod{26}.$$
Ciphertext: **AJN**

Bonus Question

(2 points) Develop a program that implements a three-stage classical cryptographic scheme composed of the following components, applied in sequence:

1. Affine Substitution Cipher
2. Transposition Cipher
3. Hill Cipher

The program shall accept a given plaintext as input, perform the above three encryption procedures in the specified order, and output the resulting ciphertext. In addition, develop the corresponding decryption program that correctly reverses the encryption process in the opposite order, so that the original plaintext can be faithfully recovered from the ciphertext.

Your submission should clearly include the following:

- the mathematical formulation and implementation of each cipher,
- the key(s) used in each stage,
- the intermediate output produced after each encryption step,
- the final ciphertext,
- the complete decryption procedure,
- and verification that the recovered plaintext matches the original input.

Please ensure that the implementation is well-structured and that each stage of the encryption and decryption process is clearly explained.

Submission Guidelines

1. Upload a zip file which contains
   - A **single PDF file** named `<student_id>.pdf` (where `<student_id>` is replaced by your actual student ID) to the new e3 platform. Present your solutions in **numerical order**, clearly labeling each subproblem.
   - Problem 1 (a) program named `<student_id>_1_a.py`
   - Problem 1 (g) program named `<student_id>_1_g.py`
   - Problem 3 program named `<student_id>_3.py`
   - Bonus problem program named `<student_id>_bonus.py`, please write the encryption and decryption functions in the same file.

Grading Policy

1. Problem 1 consists of 7 subproblems, each worth 0.5 points. Problem 2 is worth 3 points, Problem 3 is worth 3.5 points, and the bonus question is worth 2 points. So there are 12 points in total for this quiz.
2. **Late Submission Penalty:** A penalty of **0.5 points per day** will be applied for late submissions, up to a maximum of 20 days. Beyond 20 days, late submissions will be assigned zero.
3. All quizzes are **mandatory**. Failure to submit any quiz will result in an automatic failing grade for the course.